

■ RevRecover AI — 5-Minute Video Recording Teleprompter

Exact Screen Navigation, Buttons to Click & Word-for-Word Spoken Dialogue | Razorpay Buildathon

PRE-RECORDING CHECKLIST:

- **Browser Setup:** Open <http://localhost:8501> in full screen (F11 or 1920x1080).
- **FastAPI Server:** Keep running in background on port 8080 (<http://localhost:8080/docs>).
- **Screen Recorder:** Set to record browser window + crisp microphone audio (Loom, OBS, or QuickTime).
- **Pacing & Runtime:** Target duration is **4:45** (giving you a safe buffer before the 5:00 limit).

0:00 - 0:45 — The Problem & Core Value Proposition

TAB: Main Dashboard (Top)

ACTION & SCREEN:	Start at the top of http://localhost:8501 . Point your cursor at the 6 KPI metric cards (Revenue at Risk, Gross Recovered, Net Lift, Recovery Rate, PTP, Compliance Stops).
WHAT TO SAY (READ OUT LOUD):	"Hi everyone! I am presenting RevRecover AI for Track 03: AI Revenue Recovery. In Indian digital commerce, revenue loss rarely happens in a single clean step. A UPI payment degrades due to transient bank downtime, a checkout gets abandoned on the OTP screen, a recurring SaaS mandate bounces on billing day, or a B2B invoice ages past 60 days without structured follow-up. Most businesses handle this naively: they blast generic SMS spam that ruins customer trust, or they do nothing at all. We built RevRecover AI—an autonomous closed-loop multi-agent engine that detects revenue at risk, evaluates mathematical Expected Value, executes bounded recovery across WhatsApp, SMS, and Hinglish AI Voice, and proves measured money recovered with an immutable cryptographic audit ledger."
PRO TIP:	<i>Keep your delivery punchy and energetic. Stress that Indian merchants lose 5% to 15% of GMV silently!</i>

0:45 - 1:50 — "The Bar": Batch Benchmark Arena		TAB: ■ Batch Benchmark (Tab 1)
ACTION & SCREEN:	Click on the ■ Batch Benchmark tab. Leave "Composite Full Spectrum (100 Transactions)" selected. Click the purple button "Execute Batch Recovery". Watch the real-time money counter tick up. Scroll down to show the Plotly Recovery Funnel and Channel Donut.	
WHAT TO SAY (READ OUT LOUD):	<i>"Let us head straight to The Bar: proving measured money recovered across a realistic batch.</i> <i>I will select the Composite Full Spectrum benchmark of 100 transactions and click Execute Batch Recovery. Watch our real-time streaming pipeline process each transaction.</i> <i>Out of ■32.3 Lakhs at risk across e-commerce, SaaS mandates, and B2B invoices, RevRecover AI achieved a 71.8% gross recovery rate, winning back over ■23.2 Lakhs. Compared to a naive static retry policy that only recovers 22%, our multi-agent engine delivered over ■15.8 Lakhs in Net Revenue Lift after deducting all message and API contact costs.</i> <i>Below, you can see our Plotly pipeline funnel showing transactions progressing from At-Risk to Diagnosed to Recovered, and our Contextual Bandit donut routing intelligently across WhatsApp, SMS, and Silent API Retries. We can also export full audit records with one click."</i>	
PRO TIP:	<i>Scroll down smoothly while speaking so the reviewer sees the interactive Plotly funnel and channel donut.</i>	

1:50 - 2:45 — RBI Mandate Retry Sequencer & Expected Value		TAB: ■ Mandate Sequencer (Tab 3)
ACTION & SCREEN:	Click on the ■ Mandate Sequencer tab. Point out the 5-step dunning calendar on screen (Silent Retry -> T+24h SMS -> T+72h WhatsApp -> T+7d Voice -> T+14d Human Desk).	
WHAT TO SAY (READ OUT LOUD):	"Next, let us examine our RBI-Compliant Mandate Retry Sequencer. In India, recurring e-mandates and UPI Autopay must strictly adhere to the RBI circular: a maximum of 3 automated retries within 30 days, minimum 24-hour intervals, and 24-hour pre-debit notifications. RevRecover AI implements an audited 5-step dunning calendar: • Step 0: Immediate Silent API retry—zero customer intrusion. • Step 1: T+24h SMS notification with a 1-click Razorpay payment link. • Step 2: T+72h WhatsApp outreach with a smart 5% incentive. • Step 3: Day 7 Hinglish conversational voice call for high-ticket accounts. • Step 4: Day 14 formal notice escalated to a human collections desk. Crucially, every single step is bounded by our Expected Value formula: EV equals Probability of Recovery times Amount, minus Intervention Cost, minus Churn Risk times Customer LTV. If EV is zero or negative, the agent suppresses outreach automatically."	
PRO TIP:	Highlight the formula: $EV = P(\text{rec}) * \text{Amount} - \text{Cost} - P(\text{churn}) * \text{LTV}$. It demonstrates strong AI judgment!	

2:45 - 3:35 — Compliance Shield & Hard Stopping Rules		TAB: ■ Live Event (Tab 2)
ACTION & SCREEN:	Click on the ■ Live Event tab. Under Quick Scenario Presets, click "■ DND Opt-Out Test ". Click the blue button " Ingest & Trigger Autonomous Workflow ". Point out the red compliance stop banner and ZK-certificate.	
WHAT TO SAY (READ OUT LOUD):	"An autonomous agent is only as good as its safety guardrails. Let us test our Compliance Governor and Hard Stopping Rules. I will click the DND Opt-Out Test preset and ingest the event. Watch what happens immediately: our Compliance Governor detects that the customer requested opt-out under DPDP 2023 regulations. It executes an immediate hard STOP. Zero messages are dispatched, zero customer spam, and a cryptographic compliance certificate is generated. The same non-bypassable stopping rules apply to suspected fraud, active invoice disputes, and Promise-to-Pay commitments. If a customer promises to pay next Tuesday, all dunning reminders freeze completely during the grace period."	
PRO TIP:	Point your cursor at the " STOPPED_COMPLIANCE " badge and the verified DPDP compliance certificate.	

3:35 - 4:25 — Hinglish AI Voice Recovery & Mid-Call Link		TAB: ■■ Hinglish Voice (Tab 8)
ACTION & SCREEN:	Click on the ■■ Hinglish Voice tab. Click any preset dialogue, or type: "Payment fail ho gaya tha, abhi UPI se link bhej do main pay kar dunga." Click " Process Voice Turn ". Point out the agent Hinglish reply and the live payment link.	
WHAT TO SAY (READ OUT LOUD):	"Now let us look at our Hinglish Conversational Voice Recovery Agent. In the Indian market, generic English automated IVR calls get hung up within 3 seconds. Our agent conducts natural, empathetic bilingual conversations in Hinglish. When the customer explains, "Payment fail ho gaya tha, UPI link bhej do," our agent detects the positive sentiment, confirms the Promise-to-Pay commitment, and dynamically dispatches a 1-click Razorpay payment link mid-conversation so the customer can complete payment while still on the phone. This converts high-friction phone drop-offs into instant, successful UPI checkouts."	
PRO TIP:	Read the customer Hinglish quote with natural pronunciation. Reviewers will love it!	

4:25 - 5:00 — Architecture, Test Suite & Strong Closing		TAB: ■■ Audit Ledger (Tab 10)
ACTION & SCREEN:	Click on the ■■ Audit Ledger tab. Show the cryptographic SHA-256 hash table. If possible, briefly show your terminal displaying 52 passed in 5.5s .	
WHAT TO SAY (READ OUT LOUD):	"Under the hood, RevRecover AI is backed by an enterprise FastAPI backend running on port 8080 with 15 REST endpoints, an immutable SHA-256 cryptographic audit chain, and an automated test suite with 52 passing unit, integration, and end-to-end tests. We do not just identify where revenue is slipping away—we close the loop, protect customer trust, and win the money back. Thank you!"	
PRO TIP:	End with confidence. Check your recording timer to ensure it finishes between 4:30 and 4:50.	

■ Application Form Quick-Paste Reference

Keep this handy when submitting at forms.gle/d9r2gvxp8cmoZhon9:

Field	Exact Value to Paste
Track	Track 03 — AI Revenue Recovery
Project Name	RevRecover AI
GitHub URL	https://github.com/Praveen-ing/Razorpay-buildathon
What It Solves	Indian merchants silently bleed 5% to 15% of GMV to fragmented payment failures, abandoned checkouts, failed UPI autopay e-mandates, and overdue B2B receivables. RevRecover AI is an autonomous closed-loop multi-agent engine that ingests failure telemetry across 40+ Razorpay error codes, computes a mathematical Expected Value ($EV = P(\text{recovery}) * \text{Amount} - P(\text{churn}) * LTV$) to prevent over-contact, and executes bounded recovery across WhatsApp 1-click links, SMS, and empathetic Hinglish AI Voice calls. Every action strictly adheres to RBI e-mandate guidelines (max 3 retries, 24h intervals) and DPDP 2023 quiet hours, backed by an immutable SHA-256 cryptographic audit ledger proving measured net revenue recovered across high-volume batches.
What Broke & How You Got Out (Screened First!)	What Broke: - The Naive LLM-First Trap: Initially routing every failure to an LLM caused 2.8s latency, token costs exceeding margin on <■500 tickets, and stochastic hallucination of unauthorized discount policies. - Webhook Race Conditions: Burst webhook re-deliveries caused customers to receive duplicate SMS/WhatsApp messages within seconds. How We Got Out: - Hierarchical AI Judgment: Replaced LLMs at the routing layer with a fast (<5ms) Contextual Thompson Sampling Bandit + mathematical Expected Value formula ($EV = P * \text{Amt} - \text{Cost} - P_{\text{churn}} * LTV$). Constrained generative LLMs strictly to bilingual Hinglish objection handling. - Atomic Idempotency Store: In-memory SHA-256 idempotency ledger with atomic test-and-set locks to eliminate duplicate outreach. - Immutable Compliance Governor: Enforced RBI 24h intervals and DPDP quiet hours as hard pre-execution assertions.